Evaluating Data Quality and Fitness-for-Use in EPA Air Quality Monitoring Data

IS 537 Final Project Report
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GitHub Repository: https://github.com/smrutis3/IS537_FinalProject

1. Motivation and Research Question

Air quality data drives a lot of real decisions in the United States public health policy, Clean Air Act enforcement, exposure research. The EPA runs the Air Quality System (AQS), which pulls measurements from thousands of states, local, and tribal monitors into a single place. By the time any of this reaches an analyst, it has already gone through sensors, calibration, lab work, agency QA, and federal aggregation. Every one of those steps can introduce missing values, inconsistencies, or reporting artifacts that look like errors but are intentional. That last category is the one that gave us the most trouble.

We approached the dataset using the course idea of fitness-for-use: data quality is not a property of the data on its own, but of the data with respect to a question. Our question is:

How do underlying data quality issues in EPA AQS datasets affect the reliability of county-level pollutant comparisons in the 2025 annual concentration data?

Framing it this way changes what counts as a "problem." A column that is missing 99% of the time might be fine for our purposes. A column missing 10% of the time can be a serious issue if it happens to be a geographic identifier we need. We tried to keep this lens on every cleaning decision below, instead of chasing some abstract version of "clean data."

2. Datasets and Design

We worked with two files from the EPA AirData portal. The Annual Concentration by Monitor file (2025) has 59,154 rows and 55 columns one row per monitor-parameter-duration-standard combination. The AQS Site Metadata file (aqs_sites.csv) has 20,952 rows and 28 columns, and it covers every monitoring station EPA has on record. The two files ship together but they come from very different processes: the concentration file is downstream of a sensing and aggregation pipeline, while the sites file is more like an administrative registry. That gap is what made the integration interesting rather than a routine join.

Both files are at *https://aqs.epa.gov/aqsweb/airdata/download_files.html*. Our notebook expects *annual_conc_by_monitor_2025.csv* and *aqs_sites.csv* in the same folder and uses relative paths.

Quality Dimensions and Operationalizations

We picked five quality dimensions and tied each to a concrete metric so we could measure quality instead of just talking about it.

Dimension	Operationalization (metric)

Completeness	Percent non-null per field, separated by structural vs. content gaps
Consistency	Cross-dataset join match rate; agreement of shared fields (datum, coordinates)
Validity	Domain constraints: concentration ≥ 0; observation percent ≥ 75% (EPA threshold)
Uniqueness	Exact-row duplicates and business-key duplicates on (site, parameter, POC, duration, standard)
Coverage	Active sites per state and distribution of location settings

3. Quality Assessment and Cleaning Workflow

The workflow was profile, then detect, then repair. We tried to flag problems instead of imputing where we could, document everything, and avoid quietly changing data that had a reasonable explanation behind it. The rest of this section is organized by the type of cleaning involved, since structural, syntactic, and semantic cleaning each justify different repair choices.

3.1 Structural and Syntactic Cleaning

Two structural problems showed up right away, and either one would have caused a silent join failure if we had ignored it. The first was a naming mismatch, the same logical field is called *Site Num* in the concentration file and *Site Number* in the sites file. The second was a type mismatch - *State Code* is int64 in one file and object (string) in the other. A naïve merge on these would either fail outright or, worse, succeed for a subset of records and quietly drop the rest.

We handled both at once by building a composite site_key in the format SS-CCC-NNNN with zero-padding. This is structural cleaning in the strict sense: we did not change what any value means, just how it is represented, so that the same physical site is encoded the same way in both files. After harmonization all 2,246 active sites in the concentration file matched a record in the sites metadata. 100% join rate, no records dropped.

3.2 Semantic Cleaning

Three issues fell into semantic cleaning. We are using the language from class for these - denial constraints, quality rules, and key-based deduplication.

Negative concentrations as a denial-constraint violation. The constraint here is "Arithmetic Mean ≥ 0" because pollutants cannot exist in negative quantities. Profiling turned up 1,006 records that violate it (1.7% of the data), all in trace-metal speciation parameters like Cesium PM2.5 LC, Indium PM2.5 LC, and Silver PM2.5 LC. Values ranged from −42.25 down to about −0.000005. The reason these exist is that the analytical method subtracts a method blank from the sample reading; near the detection limit, the blank can come out higher than the sample. EPA's Method Detection Limit Procedure, Revision 2 (U.S. EPA, 2016) says directly that both positive and negative outputs are valid numerical results from the procedure. EPA's Unified Guidance (U.S. EPA, 2009) and ITRC's no detects guidance (ITRC, 2013) both say substituting

zero or half the detection limit is acceptable when the affected fraction is small (under about 10–15%). We went with zero rather than half-MDL because per-record MDL values are not in the dataset, and zero is the more conservative choice, it slightly underestimates rather than overestimates true concentration. We kept the original values in arith_mean_raw so the substitution is reversible if anyone wants to revisit it.

Observation completeness as a quality rule. EPA only treats annual statistics as fit for regulatory use when the underlying data covers at least 75% of the year (U.S. EPA, 2024). Once we wrote that down as a quality rule, we found that 89.4% of records (52,880 out of 59,154) fail it. Removing them would basically destroy the dataset, so instead we added an obs_quality flag with values "sufficient" or "insufficient." That keeps everything in the file but let's anyone downstream filter by reliability if they want which lined up with our preference for flagging over imputation.

Event Type duplicates as a key-based deduplication problem. There are zero exact-row duplicates. But once we defined a business key on (State Code, County Code, Site Num, Parameter Code, POC, Sample Duration, Pollutant Standard), 992 logical duplicates appeared. The reason is that EPA reports the same physical measurement multiple times under different Event Type labels. For example, the same PM2.5 mean shows up once with wildfire days included and once with concurred events excluded. This is intentional reporting structure, not an error. But for our use case (baseline comparison) it would silently double count those measurements in any aggregate. We deduped by ranking Event Type as No Events > Concurred Events Excluded > Events Excluded > Events Included, then keeping one record per business key. That removed 627 variants and left us with 58,527 records.

3.3 Missing Values: Mechanisms, Not Just Counts

Counting NULLs is where a quality assessment starts, not where it ends. The course distinguishes between MCAR, MAR, and MNAR mechanisms, and we tried to classify each field that mattered to us:

- **Pollutant Standard (64.3%) and exceedance counts (77–79%) — MNAR, structural.** Most parameters do not have a NAAQS, so these fields are blank by design. The missingness is determined by the parameter type itself.
- **CBSA Name (10.1%) — MAR.** This depends on site location: rural and non-metropolitan sites are not part of any Core-Based Statistical Area. The states most affected are Montana, California (rural counties), Wyoming, Kentucky, and Maine.
- **Method Name (22.5%) — MAR.** Concentrated in PM2.5 (8,124 records) and Ozone (3,540), and over-represented in California, Texas, and Pennsylvania. That pattern looks like state- or instrument-level reporting variation.
- **Zip Code in sites (48%) — MAR.** Older sites were registered before zip code was a captured attribute, so missingness tracks site age.

The MNAR-structural ones are not really missing data, they are non-applicable cells. Once we separated those from the MAR cases, our priorities shifted: we stopped trying to impute structural blanks, and we stopped describing the dataset as "64% incomplete," which would have been misleading.

3.4 Coverage: A Limit of Cleaning

Active sites per state range from 219 in California to 1 in the Virgin Islands. That is a 113× imbalance, and there is no cleaning operation that creates new monitoring stations. We documented it as a fitness-for-use constraint that limits which cross-county comparisons are meaningful in the first place.

4. Summary of Findings

Our consolidated detection-and-repair table is below, organized by quality dimension and cleaning type.

Dimension	Issue	Detection	Repair	Type
Validity	1,006 negative concentrations	Denial constraint: value ≥ 0	Set to zero (raw preserved)	Semantic
Validity	89.4% below 75% observation	Quality rule: obs ≥ 75%	Flag, do not remove	Semantic
Uniqueness	627 logical duplicates removed	Business-key dedup	Keep priority Event Type	Semantic
Consistency	Naming + type mismatch	Schema comparison	Composite key SS-CCC-NNNN	Structural
Consistency	Mixed datums	Value-frequency check	Documented (sub-meter)	Semantic
Completeness	CBSA Name (10.1%)	NULL count + state breakdown	Not repaired (MAR)	Syntactic
Completeness	Method Name (22.5%)	NULL count + parameter breakdown	Not repaired (cause unclear)	Syntactic
Coverage	113× state imbalance	Sites-per-state distribution	Documented (cannot clean)	N/A

4.1 Impact Assessment: PM2.5 County Comparisons

We picked PM2.5 (parameter code 88101) as a focused impact case. The raw file had 13,353 PM2.5 records across 636 counties. After the negative-value fix and the Event Type dedup, we were down to 12,826 records still covering the same 636 counties. The average county-mean change was −0.0005 µg/m³. Only 53 of 636 counties shifted by more than 0.001 µg/m³, and the largest single shift was 0.47 µg/m³. So, the visible repairs barely moved any county-level numbers.

The 75% observation completeness filter is a different story. Applied as a strict quality rule, only 1,901 of the 12,826 records survive, covering 127 counties, an 80% loss in geographic coverage. The cleaning

decision with the biggest downstream impact was not any of the repairs we wrote code for. It was the threshold we picked for what counts as a usable annual measurement.

5. Reflection: What Changed, What Did Not, and Why

Most of our county-level numbers did not change much after cleaning, and we think that is a meaningful finding rather than a disappointing one. The structural cleaning was binary in its effect: without it the integration fails, with it the integration works. There is no "how much" to measure. The semantic repairs to negative values and Event Type duplicates moved the numbers very little because the affected records are mostly in trace metals (the negatives) and a minority of pollutants (the event variants), neither of which dominate aggregate PM2.5 statistics. But both repairs are still necessary if you want defensible analysis. Negatives violate a physical constraint. Event-variant duplicates silently double-count. The fact that the impact is small is a sign the dataset is well-managed at the value level not a sign that the cleaning was unnecessary.

The story for fitness-for-use comes down to observation completeness. Whether a county-level comparison is trustworthy depends almost entirely on what threshold an analyst decides to use for observation percent. At 0% you get 636 counties of unreliable means. At 75% you get 127 counties of reliable means. Cleaning does not make this choice for you, and the data does not either. Documenting that tradeoff was, for us, the most important thing the project produced.

6. Problems Encountered

Three problems really shaped how the work went. First, the schema differences between the two EPA files are not documented anywhere in the AirData portal. Without doing the profiling explicitly, the merge would have come back with zero matches, and we might have spent a long time trying to figure out why. pandas do not raise an error when joining incompatible types, which is the kind of thing that reinforces why profiling must be a default step instead of an optional one.

Second, the negative concentration values looked like errors at first. If we had treated them that way, we would have been throwing out perfectly valid measurements. Reading EPA's Method Detection Limit Procedure (U.S. EPA, 2016) and the supporting guidance (U.S. EPA, 2009; ITRC, 2013) was what made it possible to interpret the values correctly. The lesson there is the obvious one: domain knowledge often makes the difference between a clean dataset and one that has been quietly distorted.

Third, the very high missingness percentages we saw initially (99.6% on non-overlapping max values, for example) looked alarming at first glance. They turned out to be structural those fields only exist for specific pollutant standards. Once we separated structural missingness from real gaps, our priorities changed completely.

7. Next Steps

A few things would extend this work usefully. Running the same workflow over multiple years would tell us whether the patterns we saw (89% below the observation threshold, the structural missingness profile, the geographic imbalance) are stable from year to year or specific to 2025. The Method Name missingness is concentrated in particular states and pollutants, which suggests reporting variation at the agency level that would be worth following up on directly with EPA. And the binary 75% cutoff is a coarse instrument; a continuous reliability weight based on observation percent would let comparisons keep more counties at the cost of having to model uncertainty explicitly.

8. Reproducibility and Artifacts

All code, intermediate outputs, and the cleaned dataset are in our GitHub repository. To reproduce: clone the repository, download *annual_conc_by_monitor_2025.csv* and *aqs_sites.csv* from the EPA AirData portal (*https://aqs.epa.gov/aqsweb/airdata/download_files.html*), put them next to the notebook, and run all cells in *IS537_FinalProject.ipynb*. The notebook uses relative paths only. The cleaned output gets written to *epa_aqs_cleaned_2025.csv* (58,527 rows × 18 columns), with the original concentration values kept in arith_mean_raw and the observation flag kept in obs_quality so all repairs stay reversible.

9. GenAI Declaration

We used Claude (Anthropic) to help structure the profiling workflow, generate code templates, and review the organization of this report. We used ChatGPT for code debugging and clarifying error messages. All code was tested and adapted by us, and all analytical decisions, interpretations, and conclusions are our own.

References

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